Some languages, such as Java, make this easy. They dictate that each outer-level class should be in its own file and that file should be named according to the name of the class. Other languages, such as Ruby, have no rules relating source files and their content. In Ruby, you're free to organize your code as you like.

That said, you'll find that some kind of consistency really helps. It will make it easier for you to navigate your own projects, and it will also help when you read (or incorporate) other people's code.

The Ruby community has largely adopted a de facto standard. In many ways, it follows the spirit of the Java model, in that each file is intended to have one top-level module or class and the name of that class is based on the name of the file. (If you use Rails, the Zeitwerk[2] auto-loader enforces this pattern. You can also use Zeitwerk to add auto-loading to your own gems.)

Looking at the anagram code, there appear to be three sections. The first twenty-five or so lines do option parsing, the next ten or so lines read and convert the dictionary, and the last few lines look up each command-line argument and report the result. Let's split our file into four parts:

- An option parser
- A class to hold the lookup table for anagrams
- A class that looks up words given on the command line
- A small command-line interface

The first three of these are effectively library files, used by the fourth. We can turn this into a gem, allowing access to anagram features for any program, plus the command-line interface. The standard gem structure is given to us by Bundler, which doesn't quite go with Bundler's other features but is a standard tool that you are extremely likely to have around.

All we need is to name the gem. As it happens, both "anagrams" and "anagram" are already taken. Let's go with "aaagmnr", which is the "signature" of anagram given the code's implementation of signatures, and is somehow not taken in the RubyGems central listing as I write this (there's a note about changing the initial git branch that we elided from the output):

```
$ bundle gem aaagmnr
Creating gem 'aaagmnr'...
MIT License enabled in config
Code of conduct enabled in config
Changelog enabled in config
Standard enabled in config
Initializing git repo in /Users/noel/projects/pragmatic/ruby5/Book/code/gems/aaagmnr
      create  aaagmnr/Gemfile
      create  aaagmnr/lib/aaagmnr.rb
      create  aaagmnr/lib/aaagmnr/version.rb
      create  aaagmnr/sig/aaagmnr.rbs
      create  aaagmnr/aaagmnr.gemspec
      create  aaagmnr/Rakefile
      create  aaagmnr/README.md
      create  aaagmnr/bin/console
      create  aaagmnr/bin/setup
```

2. https://github.com/fxn/zeitwerk